SIMULATION & MODELING STUDY

Discrete-Event Simulation of an Apparel Sewing Line

Bottleneck Identification and Line-Rebalancing Analysis of an 18-Process Sewing Line at Aduri Apparels, Modeled in Arena

18	26	150	10
Process Stations	Operators / Helpers	Entities per Run	Replications

Problem Statement & Objectives

PROBLEM STATEMENT

Adhury Apparels' 18-process sewing line was observed and time-studied to build a discrete-event simulation model in Arena. At the first station, Body Matching, cut fabric is drawn from a bundle that is always kept stocked, so the operator is never constrained by material supply.

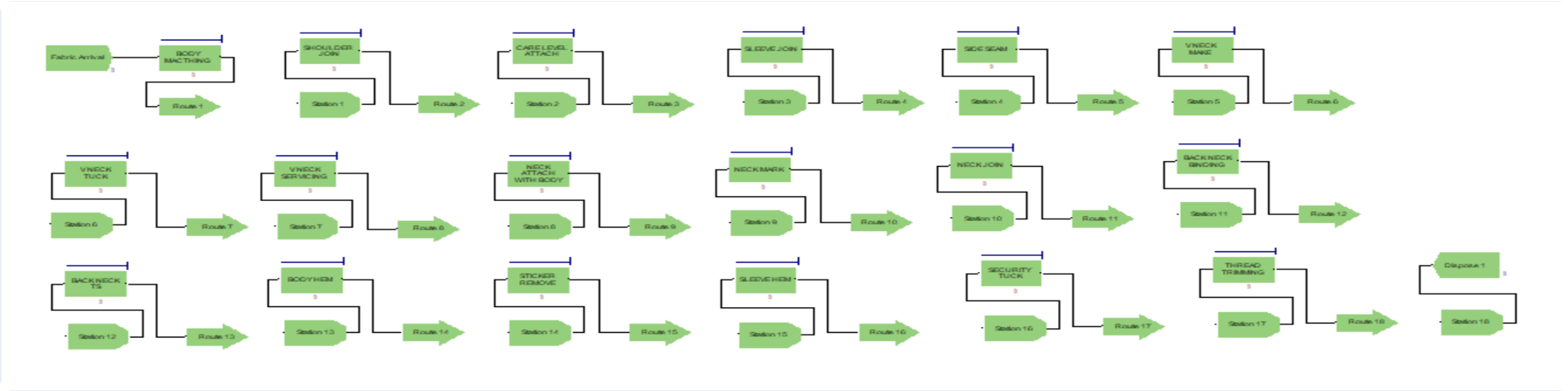
Cycle-time data — 15 observations per worker across 26 operators and helpers — was collected and fitted to probability distributions for use in the model.

18	26	15
Processes	Operators / helpers	Samples /per worker

OBJECTIVES

1. Develop an Arena model of the full 18-process sewing line
2. Simulate and visualize the line's operation end to end
3. Determine each operator's / helper's utilization
4. Identify bottleneck operations and areas for line rebalancing

Methodology



Arena model layout — Body Matching through Thread Trimming, 18 Process–Station–Route sub-models in sequence

DATA COLLECTION

15 cycle-time samples per worker across all 18 processes, fitted to Beta, Uniform, Normal, and Erlang distributions using the Input Analyzer.

MODEL STRUCTURE

Create (“Body Part Arrival”) → 18 sequential Process–Station–Route sub-models → Dispose.

PARALLEL-RESOURCE STATIONS

Sleeve Join, Side Seam, Neck Attach with Body, Neck Join, Back Neck T/S, Sleeve Hem, and Thread Trimming each use a Resource Set (Preferred Order rule) for multiple workers sharing one task.

ARRIVAL & RUN SETUP

Interarrival time set to Body Matching's own fitted distribution (12.5 + 3·BETA(2.12, 2.2) sec) to approximate the stocked Fabric within the software's entity limit. Max Arrivals = 150, 10 replications, seconds.

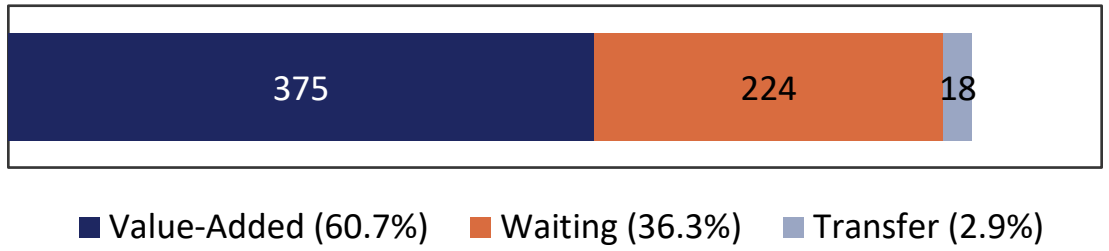
Key Findings & Discussion

150 / 150
Pieces completed per run — all 10 replications

617.8 sec
Average total cycle time per piece

2.57
Average work-in-process (pieces)

CYCLE-TIME COMPOSITION (AVG. PER PIECE)



TOP BOTTLENECKS — AVG. WAIT TIME PER PIECE

Care Level Attach (Station 2 — Savana)	72.1 sec
V-Neck Servicing (Shiuli)	56.2 sec
Shoulder Join (Nasima)	48.7 sec

All three bottlenecks are single-worker stations — despite moderate individual service times, they form the largest queues on the line, making them the clearest rebalancing candidates. Resource utilization across all 26 operators/helpers was generally low (≈ 5–7%), a direct result of the interarrival pacing used.

Impacts, Limitations & Future Work

1 IMPACTS

- Pinpointed three concrete rebalancing candidates: Care Level Attach, V-Neck Servicing, and Shoulder Join
- Demonstrated a working, data-driven simulation framework for evaluating line performance without requiring cost data

2 LIMITATIONS

- Arena's 150-entity license cap blocked a bottleneck-paced (≈28–30 sec) or batch/bundle arrival process — both raised a “Maximum entities exceeded” error before steady state
- No labor cost data was provided, so cost-based analysis (labor cost, cost per unit) could not be performed
- Based on single-shift observation with only 15 cycle-time samples per worker, limiting statistical robustness

3 FUTURE WORK

- Collect actual labor cost data to enable cost-based optimization analysis
- Test the effect of adding a second worker at Care Level Attach, V-Neck Servicing, and/or Shoulder Join
- Collect a larger number of cycle-time samples per operation to improve distribution-fitting accuracy